Installation & Deployment Guide

Complete Step-by-Step Setup for Windows and Linux

1. System Prerequisites

• OS: Windows 10/11, Ubuntu 22.04+, or macOS

• Python: Python 3.11 or 3.12 (64-bit)

• RAM: 16 GB minimum (32 GB recommended for GPU acceleration)

• Storage: 10 GB free disk space for local models and vector database

• Ollama: Local Ollama daemon installed (https://ollama.ai)

2. Windows Installation Steps

Step 1: Open PowerShell as Administrator and create a virtual environment:

python -m venv venv

.\venv\Scripts\Activate.ps1

Step 2: Install required Python dependencies:

pip install -r 3_Code\requirements.txt

Step 3: Install and configure Ollama:

Download and run OllamaSetup.exe from https://ollama.ai

Run in terminal: ollama pull qwen3.5:9b (or qwen2.5:7b)

Step 4: Generate the synthetic demonstration PDF:

python 3_Code\sample_data\generate_sample_hld.py 2_Input_Data\Sample_AUTOSAR_HLD.pdf

Step 5: Launch the Streamlit application:

cd 3_Code

streamlit run app.py

3. Linux / macOS Installation Steps

Step 1: Create and activate virtual environment:

python3 -m venv venv && source venv/bin/activate

Step 2: Install dependencies:

pip install -r 3_Code/requirements.txt

Step 3: Start Ollama service and pull model:

ollama pull qwen3.5:9b

Step 4: Launch application:

cd 3_Code && streamlit run app.py